

Biodiesel in South American countries: A review on policies, stages of development and imminent competition with hydrotreated vegetable oil

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ABSTRACT

Economic and population growth have led to rapidly growing fuel consumption in South America. In order to supply this increasing demand while also using environmentally friendly and locally available feedstock, several countries have implemented the use of biofuels, in which biodiesel deserves special attention. While this alternative source of energy may reduce dependence on refining or importing fossil diesel, it also requires well-established policies and logistics for local industries to compete and succeed in producing a satisfactory quantity of biodiesel. This review compares the effectiveness of implemented policies in South American countries, providing a better understanding on cases of success and failure on the road to making biodiesel a regularly found fuel in the local transportation network. This assessment also clearly identifies different stages of development, pointing out that countries from the same region can have totally different outcomes according to political and market aspects. The main factors associated to cases of success, represented by Brazil, Argentina and Uruguay, were early implemented regulation policies, the solidity of local soybean industry and response to foreign competition. Finally, Hydrotreated Vegetable Oil (HVO) is a new and efficient green fuel used in Diesel engines and will likely present additional challenges for this sector, potentially replacing both diesel and biodiesel in the fuel network, and posing a challenge to even the most stable biodiesel industries, along with the potential vehicle electrification.

1. Introduction

Both population growth and economic development have led to an increase in energy consumption in South American countries, which extends from power generation [1] to the transportation sector [2], forcing nations to look for new sources of energy. In fact, the number of vehicles in South America has significantly increased [3], in such a way that the demand for fuels in the region have strongly increased over this same period. Even though several of these countries rely on importing or producing fossil fuels like gasoline and diesel, strong and diversified agricultural production in the region [4] and lacks in crude oil refining technology [5] make biofuels like biodiesel an attractive alternative for certain regions.

In this context, several South American countries have started implementing policies to regulate and encourage biofuel production, each focusing on its particular scenario. The particularities of each

country and corresponding biodiesel regulations – which will be discussed here – depend both on internal and external factors, and can be used to open national markets by encouraging imports or protecting national producers from foreign competition. Variations in regulations have also been observed overtime, as nations should be able to properly react to market changes to keep the biodiesel production and the demand for fuels under balance.

Although several other studies on biodiesel policies can be found for specific countries, regions or historical periods, the goal of this review was to provide a summary, particularities and policies (or lack thereof) for all South American countries. This study chose not to focus only on major regional producers (i.e. Brazil and Argentina), but also to include at least a quick overview for biodiesel perspective in every South American country. Since geographic location is the main link between the nations studied, biodiesel perspectives in French Guiana – which is not technically a country – were also reviewed.

The main idea of performing a systematic review on this topic is that

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List of abbreviations:

ANCAP	Uruguayan National Administration of Fuel, Alcohol and Portland
ANP	Brazilian National Agency for Petroleum, Natural Gas and Biofuels
BRL	Brazilian Real (currency unit)
CI	Compression ignition
COP21	21st Conference of the Parties
CRE	French Energy Regulation Commission
DICAPAR	Associated Fuel Distributors of Paraguay
ECLAC	United Nation's Economic Commission for Latin America and the Caribbean
EPA	U. S. Environmental Protection Agency
GHG	Greenhouse gases
HC	Hydrocarbons
HVO	Hydrotreated Vegetable Oil

IBP	Brazilian Institute of Petroleum and Gas
LPG	Liquefied petroleum gas
NDCs	Nationally Determined Contributions
NGOs	Non-governmental organizations
NEXBTL™	Neste Renewable Diesel
OPEC	Organization of the Petroleum Exporting Countries
PETROPAR	Petróleos Paraguayos
PM	Particulate Matter
PNPB	Brazilian National Program for Production and Use of Biodiesel
RED	Renewable European Directives
RFS	Renewable Fuel Standard
SPK	Synthetic Paraffinic Kerosene
THC	Total Hydrocarbon
Ubrabio	Brazilian Union of Biodiesel and Biojetfuel
VAT	Value Added Tax
YPFB	Yacimientos Petrolíferos Fiscales Bolivianos

understanding the rights and wrongs in the biodiesel industry of South American countries can help identify potential improvements and policy changes in future decisions, while also allowing comparisons with countries from other regions, as shown in previously published studies [6]. In this sense, several variables, that may or may not belong to the political scope, can be assessed simultaneously in order to create a clear understanding of what favors or does not favor the development and stabilization of national biodiesel industries.

This review presents a critical evaluation on biodiesel industries in the South American region by highlighting and discussing real cases of success and failure, as well as their relation with political, geographical and historical aspects. Although smaller, non-biodiesel producing countries are mentioned here, abundant data naturally gives more variables for discussions when analyzing major regional producers, like Brazil and Argentina. In order to differentiate various types of renewable diesel fuels, the term “biodiesel” in the present manuscript refers exclusively to alkyl esters produced via transesterification, which are the only renewable fuel currently used in large scale for CI engines in South America.

2. Biodiesel in South America

The development of biodiesel in each country strongly depends on its market, the ability to produce/import feedstock or biodiesel and, certainly the most important, the development of incentive policies that are able to respond to regional particularities and market changes. As will be discussed through practical examples, these policies, when properly applied, have the potential to both control the implementation of biodiesel in the market and protect new fragile local industries against bigger well-established industries that can pose threats in the form of unfair competition. Another important factor is when policies first started being implemented, which is directly related to how mature the national biodiesel industry is and how stable is the market in terms of long-term use of this biofuel in the transportation sector.

Several aspects influence the interest in developing a biodiesel industry in South America, among which the large feedstock availability can be highlighted [2]. However, controlling GHG emissions is probably the main driver behind biofuel policies in general, justifying incentives and influencing regulations in the sector. In this sense, several South American countries have presented strategies for reaching carbon neutrality by 2050, many of them betting on electrification of vehicles as a way towards a cleaner transportation system. This option, however, may not be technically viable over the next few years, a fact that has not been properly discussed in any of South American NDCs (except for Chile [7]). Recognizing that biofuels can represent an option for the

transition towards a reduction in GHG emissions while full electrification is not possible [8], biodiesel has been used in several countries over the years, and even countries with no history of biodiesel production started implementing recent regulations to produce this and more advanced biofuels [9].

In this section, the history and development of biodiesel regulation are individually discussed for every South American country.

2.1. Argentina

The first Argentinean legislation regarding biodiesel blends with diesel was released in 2006 via Law 26.093. This law introduced concepts and attributions, defining institutions that should be responsible for regulating the biofuel market in the country, including requirements for producing and commercializing biofuels, quality standards and blends with traditional fossil fuels [10]. Initially, the minimum biodiesel blend was set to 5% and became mandatory in 2010. Responsible authorities could decrease or increase this reference value according to market and national production variations. These initial attributions later regulated and detailed by Decree 109/2007, which decided that the Secretary of Energy, belonging to the Argentinean Ministry for Federal Planning, Public Investments and Services, should be the “authority of application” of this law [11].

The biodiesel industry in Argentina is almost exclusively based on soybean oil, since this is one of the strongest sectors of the national industry. Soybean plantations occupy a large fraction of grain cultivated areas and are concentrated in central provinces [12], which host almost all biodiesel production facilities currently operating [13]. Biodiesel production in Argentina is also very dependent on exports, since most of this biofuel is shipped to other countries, like the Netherlands (54.9%), Malta (31.3%), Belgium (6.4%) and Canada (6.1%), which were the main buyers in 2018 [14]. Since crude glycerol is a direct byproduct of biodiesel production via transesterification [15], Argentina also exports this to several other countries, mainly China (55.5%) and the Netherlands (7.8%), and over 30% is sent to other undisclosed locations. Moreover, biodiesel export records have been kept since 2007, before national consumption became mandatory, further highlighting the importance of Argentinean biodiesel industry exports [14].

Annual biodiesel trade balances in Argentina again highlights the major role of exports in this sector. As a result, the national production has been significantly higher than consumption over the years, and imports were rare. Even though Argentina chooses to export its surplus biodiesel instead of increasing its use within the country, it is clear that this policy ensures high stability from an internal consumption standpoint. This can be confirmed by the trade balance from 2015 and 2019

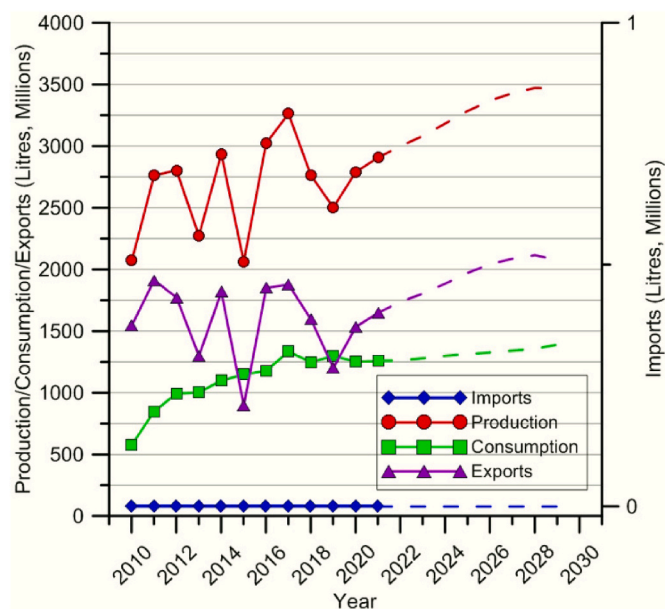


Fig. 1. Annual balance of the biodiesel industry in Argentina [16].

(Fig. 1): when production significantly dropped, Argentinean biodiesel exports also decreased, but consumption was kept constant ensuring that the diesel-biodiesel mixture could be maintained.

As foreseen in the original law, the mandatory minimum blend of biodiesel was later modified, increasing biodiesel participation in the internal market by setting its fraction to 10% volume [17]. The fact that the Argentinean biodiesel can rely on both internal and external markets adds more stability to the national industry, allowing the government to boost internal consumption or exports in response to market or political variations. As an example, recent projects presented by the national senate proposed increasing national biodiesel blends from 10% to 20%, arguing that recent barriers posed by the United States were harming Argentinean biodiesel exports [18,19]. The fact that a sudden increase from 10% to 20% is even considered – despite the technical barriers of changing the diesel's composition suddenly – highlights the flexibility of the Argentinean biodiesel market, in such a way that the lack of exports can be overcompensated by internal consumption and *vice-versa*.

Due to the Covid-19 pandemic, Argentina temporarily reduced its biodiesel production, as local and export demands decreased. This negative scenario has led to the lowest production levels in the last decade, at 1.8 billion liters less in 2020 [20]. More recently, the Argentinean law that regulates biodiesel blends was renewed, making the Secretary of Energy of the Argentinean Ministry of Economy responsible for changing the biodiesel fraction in the commercial diesel according to price and market variations, respecting a minimum of 3%. This regulation, which considers as biofuels only bioethanol and biodiesel, also added a subtle change relative to the 2006 law by applying effects only to biofuels produced in the country using national feedstock, providing additional stability for local producers [21].

2.2. Bolivia

Bolivia has been dependent on liquid fuel imports for a long time. In fact, its diesel imports have grown from 556,449 tons in 2010 to 1,213,547 tons in 2018, indicating that the increase in national fuel demand has forced the country to spend more money on buying this product from abroad [22]. Most of this diesel comes from nearby countries, such as Argentina, Chile, Paraguay and Peru. In 2018, over 46% of the diesel imports came from Argentina, followed by Chile (30.8%) and Peru (13.7%). Lower quantities were also imported from non-South American countries, such as Swaziland, Switzerland, and

Saudi Arabia [23].

At the same time, Bolivia is one of the major soybean oil exporters in the world, with around 377 thousand tons exported in 2018. Also, a significant amount of sunflower oil is exported [24,25]. Considering this current scenario and aiming to gradually reduce imports, the Bolivian government recently created a normative framework for production, storage, transportation, commercialization and mixing of vegetable based additives, whereby biodiesel is also included [26]. In this sense, the Bolivian Ministry of Hydrocarbons has announced in March 2019 an investment of 2 billion dollars and an increment of 250,000 ha to produce 100 million liters of soybean oil to be used for biodiesel production [27]. This change towards a less fossil-dependent energy matrix, however, has led to criticism by part of the local media given the environmental impacts associated with deforestation and the relatively minor impacts on the overall fuel import balance (around 8% of the imported diesel based on data from 2018).

Despite only recently moving towards biodiesel production, Bolivia is already planning to produce a more modern biofuel: HVO, also known as “green diesel” or “renewable diesel”. In March 2021, the Bolivian government, in conjunction with the state hydrocarbons company YPFB announced plans to build a new plant for producing green diesel in the “Guillermo Elder Bell de Santa Cruz” refinery. Bolivia expects to invest about 250 million dollars to produce 9000 barrels of HVO per day, using vegetable oils, used cooking oil and animal fats as feedstock. By the second half of 2021, a company will be selected to assume the project, with production to start in 2024 [9].

2.3. Brazil

Brazil is the largest country in South America in terms of population and area, and already has a long history of using renewable sources of energy, both for power generation [28] and for transportation [29]. This country is currently one of the major biodiesel producers in the world, being placed second in this biofuel production in 2018, losing only to the United States [30]. In the same year, Brazil's biodiesel production represented more than 13% of all biodiesel production worldwide [31,32], and the rise of new energy policies indicates that these numbers should increase even further.

The story of biodiesel in the Brazilian energy matrix begun in 2003, via a decree that selected a work group to perform studies regarding the viability of using biodiesel in the country [33]. These efforts were later made official at the end of the following year via the National Program for Production and Use of Biodiesel (PNPB). This program established the basis for producing and commercializing biodiesel in the country, taking several particular national aspects into account, such as regional feedstock availability, the influence of familiar agriculture and competitiveness with the traditional fossil diesel [34]. Since then, several policy changes were implemented, and the biodiesel fraction in commercial diesel fuel gradually increased from a non-mandatory value of 2% in 2004 [35] to a mandatory portion of 13% in 2021. This fraction is expected to increase even further, up to 15% by 2023 [36].

A more recent Brazilian governmental program worth mentioning is the RenovaBIO. This National Biofuel Policy (Law No. 13.576/17) evaluates and proposes improvements to new policies and regulatory aspects for better using biofuels in Brazil [37,38], in order to promote adequate expansion of biofuels and to help the country achieve GHG reduction by increasing the presence of renewable biofuels in the energy matrix [39]. This program foresees the attribution of different grades to each biofuel producer, with the highest grades being given to those that produce the highest amount of fuel energy with the lowest CO₂ emissions along its life cycle [39].

Additionally, this national policy provides carbon credits by biofuel producers certified by the RenovaBIO program, based on their energy-environmental efficiency rating. In this sense, the Brazilian carbon credit market, via biofuel producing companies, marketed over 650 thousands BRL at an average price of 43.66 BRL per carbon credit [40].

With the RenovaBIO program, a 10.1% reduction in CO₂ emissions in Brazilian fuel matrix is expected between 2018 and 2028 [41], which should result in investments in the Brazilian biofuels sector close to one trillion BRL by 2030 [42].

Most of the Brazilian biodiesel industry rely on soybean oil, whose contribution in the national production remained between 68 and 82% since 2008 [43]. This is not surprising since Brazil is expected to produce 124.5 million metric tons of soybeans in the 2019/2020 harvest, which should account for 36.8% of global production for the same period [44]. However, smaller biodiesel portions are also produced using animal fats, waste frying oil and cottonseed oil [43]. The strong dependence of Brazilian biodiesel on soybean oil, however, is a double-edged sword. If on one hand the reliance on the well-established national soybean market was essential to support the early biodiesel industry in the country, on the other hand it suffocated the potential use of other oilseeds in large scale, thwarting the idea that the biodiesel production could effectively boost family farming in less favored regions where soybean is not widely produced. This led to a quick end of the so-called Social Fuel Stamp, a government program that aimed to promote the inclusion of family agriculture into the national fuel chain [45]. A more detailed discussion on soybean dominance in the Brazilian biodiesel market over the failure of national “social fuel” programs can be found in De Oliveira et al. [45] and Hall et al. [46].

The most recent challenge faced by the national biodiesel industry has been the coronavirus pandemic. Despite the maturity of Brazilian soybean/biodiesel production system, the pandemic has led to a soybean scarcity, increasing prices and compromising the biodiesel production. As a result, the Brazilian National Agency for Petroleum, Natural Gas and Biofuels (ANP), which is responsible for regulating the biodiesel market in the country, temporarily decreased the mandatory biodiesel fraction in commercial diesel from 12% to 11% in the November/December 2020 period [47]. In April 2021, the biodiesel fraction was once again reduced during the 79th Biodiesel Auction, this time from 13% to 10% [48]. The current scenario also led to a request from the Brazilian Institute of Petroleum and Gas (IBP) for changing current regulations, among which the possibility of biodiesel imports was suggested [49]. These demands, however, are relatively new, and how national regulatory system will respond remains uncertain.

The biodiesel balance in the Brazilian scenario is presented by Fig. 2. In opposition to policies implemented in Argentina, Brazil favors internal consumption of biodiesel over exportation, although there are

significant export levels. Importation of this fuel was only present in the early years of the industry, with no significant amounts of biodiesel being imported since 2012. In general, Brazil seems to consume virtually all the volume that is produced nationally, which allows the country to keep a relatively high fraction of biodiesel in the commercial diesel blends. Even though this exposes the national fuel matrix to low production periods, like the instability caused by the coronavirus pandemic, Brazil was able to keep a production high enough to ensure that biodiesel fractions were not greatly affected even under unfavorable conditions.

Despite similar isolated episodes, the Brazilian biodiesel industry is currently stable and solid, and generally capable of responding to variations in the market demand over time. In the next few years, however, biodiesel is expected to face a local competition from Hydrotreated Vegetable Oil (HVO). This renewable diesel, which can also be produced from feedstock such as soybean oil, already received an early regulation by ANP [50], and industrial-scale tests by state company Petrobras were capable of yielding 40 million liters of co-processed diesel-HVO [51]. The choice for a co-processed fuel (5% HVO) instead of its pure version aims to decrease both production time and initial costs. Moreover, the mixture leaves the refinery ready for blending with enough biodiesel to meet the renewable fraction defined by law [52]. More recently, the first regulation on renewable diesel fuels was released by the Brazilian Ministry of Energy and Mines through ANP Resolution N° 842, which included HVO, Fischer-Tropsch and other diesel-like fuels under the “green diesel” label [53]. In the near future, complete regulation of HVO is expected to indicate whether this fuel will receive political incentives or not, defining the level of competition to be faced by the biodiesel industry in Brazil.

2.4. Chile

Chile has the highest primary energy [54] and electricity [55] consumption per capita in South America. It has an energy matrix mainly based on fossil fuels (66%) and biomass (25%) [56], being extremely dependent on non-renewable sources of energy. About 85% of total domestic energy supply in Chile comes from imports [57], placing Chile in a vulnerable position to world market price variations [58]. In this sense, Chile has been developing energy policies to promote a more diversified energy matrix by using renewable energy sources [59], and the production of biofuels could be an alternative for strengthening national energy production in their energy matrix [60].

Despite this, there is limited potential for producing biodiesel from first generation feedstock in Chile, since the country has as obstacle the low availability of agricultural soils [61,62]. Among the options for the production of biodiesel in Chile, sunflower and rapeseed crops stand out as having the greatest potential of production [63,64]. In general, the Chilean government has focused on the production of second-generation biodiesel from microalgae [62] or by using cooking oil and animal fat [65].

There are few policies that encourage the production of biodiesel in Chile; and only Circular N° 30 from May 16th, 2007 [66], exempts biofuels from fuel taxes, but does not exempt them from the Value Added Taxes (VATs) [67]. In addition, there is no regulatory framework in Chile that requires the use of biodiesel in commercial diesel for urban mobility, except for the Decree N° 11 of 2008, that established voluntary use of 2% or 5%, on a volumetric basis, of biodiesel in blends with diesel [65,68].

Román-Figueroa and Paneque [69] highlighted the need for the Chilean government to promote policies that can encourage investments in biofuels, which could be capable of respecting the environment and the society. The authors suggested using second-generation raw materials, in order to take advantage of degraded areas that are not used for agricultural or forestry purposes.

Therefore, there is no significant production of biodiesel in Chile [70], and there is no expectation for large-scale production in the near

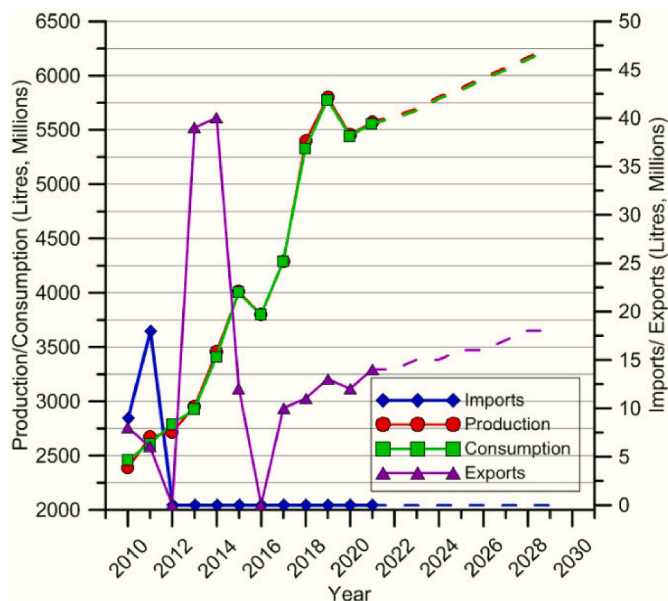


Fig. 2. Annual balance of the biodiesel industry in Brazil [16].

future. Despite this, Chile has an ambitious strategy for H₂ production for industrial and urban mobility uses [71,72].

2.5. Colombia

In Colombia, national biofuel policies were introduced back in 2002 by removing the value-added taxes on biofuels and by withdrawing global tax charges. The first legal instrument to promote biodiesel in the country was Law N° 939/2004, whereby the Colombian Ministry of Mines and Energy set quality standards, blending mandates and regulatory measures for establishing the biodiesel in the Colombian market and energy matrix [73,74].

These government actions first addressed the development of sugarcane and palm oil industries, in order to support additional revenue for farm producers. However, along with economic goal, the idea for energy matrix diversification emerged together with the need to decrease dependency on fossil fuels. More recently, Colombian authorities added to their line of action the goal of introducing environmentally friendly fuels to reduce GHG emissions and to help decrease pollution levels in major cities, and to meet Colombia's commitments under the 2015 Paris Agreement (COP21). The B12 mandate was introduced in September 2019, which was the highest blend ever established by law in the country. Unfortunately, this mandate was not successful by December 2019, since three months after B12 was released, the blend was reduced to B10, due to low palm oil production, which is the main feedstock used in Colombia. In December of 2017, a bill that creates taxes for fossil fuels based on the release of carbon dioxide (CO₂) factor for each fuel type was passed as law in Colombia. The biofuels are exempted from this tax, also known as Green Tax, but for gasoline and diesel, the taxation accounts for 139.50 and 156.60 Colombian pesos per gallon, respectively [74].

The Colombian Ministry of Mines and Energy established methodologies for calculating biofuels prices based on opportunity cost from using palm oil and sugarcane to supply other markets. The methodology for calculating the biodiesel price is based on three resolutions: N° 181,780 from 2005, N° 181,966 from 2011 and N° 181,489 from 2012. These resolutions established a range for a ceiling price based on the import parity price for fossil diesel, weighted by technical factors, along with a price floor defined based on the export parity price for palm oil, according to prices set by the Rotterdam market, adjusted using logistics and technical parameters. The reference costs for biodiesel might change across the country due to expenses of distribution for each region [74].

Not different from the rest of the world, the price of biodiesel in Colombia is higher than fossil diesel, so blends are more competitive for expanding the biodiesel market. One goal that has been pursued in Colombia is reducing biodiesel costs using feedstock with lower prices. For example, in 2019 palm oil production decreased 6% relative to 2018, because lower prices discouraged harvesting investments. The Covid-19 pandemic also directly impacted fuel demand in Colombia. Pre-pandemic consumption was 6,500 million liters of diesel-like fuels in the year. However, due to lockdown measures, the demand for diesel decreased 50% in April, and projection was that, in the end of 2020, total fuel demand would decrease 5% compared to the previous year. Considering biodiesel only, expectations are for the decrease of 555 million liters in 2020, a decrease of 8% from 2019. Fig. 3 presents the balance for biodiesel in Colombia since 2010. Similarly to Brazil, Colombia consumes virtually all of its biodiesel production, but no exports or imports have been reported. The Colombian producers could not ensure stable biodiesel consumption patterns over the years, with significant production and consumption drops in the 2015–2017. While Argentina kept its consumption constant in similar low-production periods by decreasing exports, Colombia does not produce surplus for export, so their national market is extremely vulnerable to periods of scarcity.

Currently there are 12 operational biodiesel facilities in Colombia using palm oil, and only one of these produces biodiesel from second-

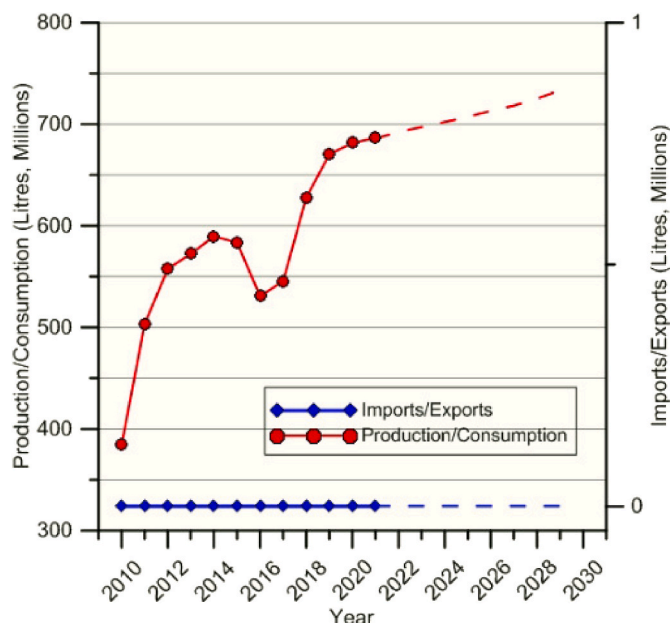


Fig. 3. Annual balance of the biodiesel industry in Colombia [16].

generation feedstock (cooking oil) in small scale, which could be a problem. The study of Colmenares-Quintero et al. [75] identified that although there is available land for the expansion of biodiesel production, the distance from these lands to refineries represents a problem and also the infrastructure currently available, both factors that increase costs and jeopardize biodiesel expansion in a national scale. Colombian biodiesel production is strictly for internal consumption, since the bio-refineries are running in one third of full capacity and there are no requests for neither exports or imports. The importation of biodiesel was authorized in 2017, however prices were not attractive. For exportation, the problem seems to be related to environmental issues from the two major purchasers, the European Union and the United States of America [73,74].

In Europe, Renewable European Directives (RED) II classified palm oil biodiesel as a biofuel with high risks for indirect land use, and limited its consumption in Europe and intends to phase it out by 2030. The problem with the United States is that The US Environmental Protection Agency (EPA) has not completed the life cycle assessment for any of the Colombian facilities. It was also reported that the biodiesel from Colombia did not meet the Renewable Fuel Standard (RFS) of the United States, and, for these reasons, Colombian biofuel cannot generate Renewable Identification Numbers, which is mandatory for EPA approval [74].

Currently, the transition from biodiesel towards new biofuels for diesel engines, like HVO, is not officially underway. There are no current official regulations, although one of the few mentions of this fuel in the country was in an early study from Colombian EAFIT University, which carried out a case study for producing pure and co-processed HVO in the Cartagena biorefinery, concluding that the project could be economically viable if green diesel received the same benefits as biodiesel. The same study, however, emphasizes that the Colombian biofuel regulations do not explicitly include HVO as biodiesel, so improvements to national policies would be necessary to increase the viability for similar projects [76].

2.6. Ecuador

Despite being capable of producing more than twice as much oil per day as it consumes [77], Ecuador can only refine about one third of its production [54]. Therefore, Ecuador exports crude oil [78], and imports

gasoline, diesel and LPG in order to supply the domestic demand [79, 80]. Additionally, Espinoza et al. (2019) [81] and Larrea et al. (2017) [82], warned that drops in oil reserves may cause the country to stop exporting oil, becoming an importer by the 2030s. In this sense, biodiesel production could be an alternative for the country to reduce fossil fuels consumption [5].

The first decree related to biodiesel production in Ecuador was Executive Decree No. 2332 of 2004, which sought to start implementing a market for commercializing biofuels in the country [83,84]. Nonetheless, this decree merely established a milestone for promoting biodiesel production in Ecuador [85]. Only in a new decree, published on September 17th, 2012 (Executive Decree No. 1303), the government decided to add a 5% fraction of biodiesel in blends with diesel, and to gradually increase this value to 10% biodiesel mixtures [86,87].

Industrial biodiesel production in Ecuador comes mainly from African palm (*Elaeis guineensis*) [88,89], with approximately 280,000 cultivated hectares [90]. Other alternatives for biodiesel production being studied/proposed are castor bean (*Ricinus communis*), cacaoy fruits (*Caryodendron orinocense*), Barbados nuts (*Jatropha curcas*) and micro algae [91,92].

The only large-scale biodiesel producer in Ecuador is La Fabril, with an installed capacity of 170 million liters per year [93,94]. On the other hand, in order to accomplish the executive decree that propose the addition of 5% biodiesel to diesel, it is estimated that annual production would need to be 240 million liters, while La Fabril would only be able to cover 30% of this demand, even for scenarios where biodiesel is not exported [93].

Therefore, it can be said that biodiesel production in Ecuador is incipient [70], with growth potential, especially given large increases in African palm cultivation in recent years [95]. A leap in biodiesel production is expected in the near future, making Executive Decree No. 2332 feasible until biodiesel production can ensure diesel blends containing 10% biodiesel [93].

2.7. French Guiana

Unlike other nations mentioned in previous topics, French Guiana is not a country, but rather a French overseas territory. Therefore, this territory responds to regulations defined by France and the European Union, which may in certain cases, create information gaps between where decisions are made and where effects are observed [96]. Despite technically being a European region, French Guiana is located in South America (bordering Brazil and Suriname) [97] and therefore will be briefly discussed here.

The French Guinean transport and energy matrix is strongly dependent on external supply and on the use of fossil-based technologies. There is a remarkable difference between coastal and inland regions in the territory, with the only electricity network supplying the coastline while fast-growing inland populations strongly depend on fossil fuels [97]. Given the major penetration of fossil diesel in the territory, the more urgent problems faced by their energy sector and the fact that neither the last “Pluriannual Energy Programming” [98] nor the French Energy Regulation Commission (CRE) [99] mention intentions for local biodiesel production in the near future, it is safe to assume that no significant amount of biodiesel is expected to be produced in French Guiana in the next few years.

2.8. Guyana

Guyana's economy is mostly based on primary resources, like mining and agriculture, and this scenario is not expected to change in the near future, mainly due to lacks in technology and infrastructure [100]. Guyanese agriculture has traditionally focused on rice and sugar plantations, but a transition is underway to include other products and reduce dependence on importations. In this context, maize, traditionally used as animal feed, has been given special attention by the government

[101].

Currently, Guyana does not have any law that regulates biofuel production, commercialization or blending with diesel, and local biodiesel production potential has been overlooked by several reports and articles. In a 2007 report prepared by the United Nation's Economic Commission for Latin America and the Caribbean (ECLAC) regarding the biofuels potential in Guyana, biodiesel is not mentioned except for conceptual purposes [102]. Similarly, a 2019 government report on sustainable development only briefly mentioned biodiesel as a potential energy source from waste [103]. The same was true in the 2020 report by the Guyanese Department of Environment aiming to discuss possible pathways for a greener energy matrix in the country [104].

Lacks in regulation and proper technology make large-scale biodiesel production in Guyana uncertain, and only small-scale, research-level projects can be found in the country. One example is the Institute of Applied Science and Technology, at the University of Guyana, which produces biodiesel from waste vegetable oil and supply almost all of the institute's fuel demand [105]. For bigger projects, Guyana will depend on foreign investment and technology, which can be achieved through Brazilian and other regional partners [106], making possible – yet not certain – the future development of a national biodiesel industry.

2.9. Paraguay

In 2005, Paraguay took its first steps towards biodiesel production in the country via Law N° 2748/05. This law introduced definitions and indicated future benefits and advantages from projects aiming the production of biofuels, opening paths for the yet-to-be-developed biodiesel industry in the country [107]. In the following year, Decree N° 7412/2006 was created to further regulate requirements and benefits described in Law N° 2748/05 [108].

The actual implementation of biodiesel in the Paraguayan fuel matrix, however, only came in 2007, when Resolution N° 235, from the Paraguayan Ministry of Industry and Commerce, stated that commercial diesel should contain a minimum of 1% biodiesel that same year, and that this should gradually be increased to 3% in 2008, and 5% in 2009 [109]. A new resolution, however, was published in 2009 setting this fraction to 1% once again. This backstep was justified in the resolution itself, arguing that national biodiesel production would only be able to supply enough fuel for mixtures as low as 1.7% [110]. A possible explanation for the inability of Paraguayan biodiesel producers to keep up with national demand and local regulations may be excess focus on feedstock flexibility, rather than on output stability. In fact, several biodiesel industries in the country operate based on feedstock costs and availability, several of them using animal fats in addition to vegetable oils that are economically viable [111].

Almost a decade later, at the end of 2018, Resolution N° 1253/2018 increased the biodiesel fraction back to 2% in Paraguayan Type III diesel, starting in January 2019 and arguing that national biofuel producers had pointed “no inconveniences” in terms of production capacity [112]. This increase was soon postponed to March [113] and again to April of the same year [114], following requests from the Associated Fuel Distributors of Paraguay (DICAPAR), and the Petr leos Paraguayos (PETROPAR) company. The actual implementation of the mandatory 2% fraction came only with Resolution N° 1085, published in September 2019, from the Paraguayan Ministry of Industry and Commerce [115].

Difficulties faced by Paraguay regarding its own fuel regulations can be observed in its biodiesel balance (Fig. 4). Similar to Colombia, Paraguay did not report any biodiesel imports or exports. Instead, the country relies on its own production capacity to partially supply the demand for biodiesel, and no surplus is generated in order to guarantee a stable consumption. Even though the consumption curve seems relatively stable, national production can only supply a small fraction of what was initially expected, taking over a decade to produce enough fuel to cover a 2% fraction in commercial diesel. In this sense, this small production and slow growth explain the backsteps and postponed

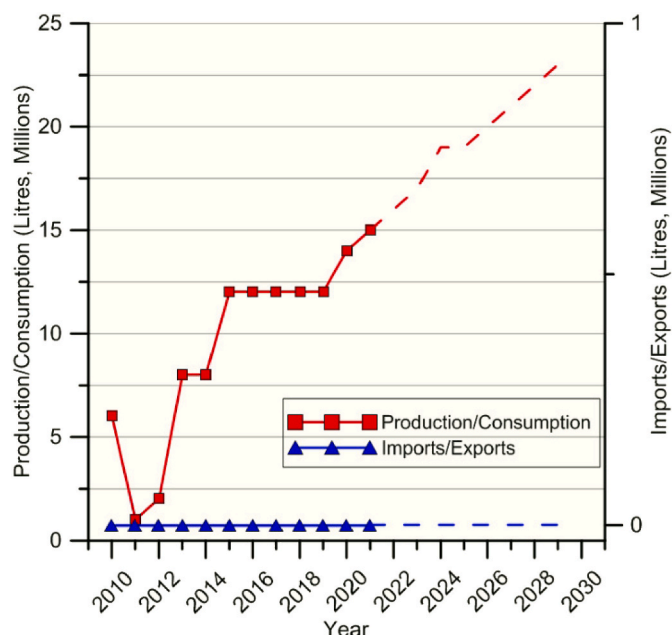


Fig. 4. Annual balance of the biodiesel industry in Paraguay [16].

implementations in Paraguayan biodiesel policies over the years.

In parallel to its national biodiesel industry, Paraguay is also expected to soon introduce a new biofuel in its energy matrix. The Omega Green biorefinery, developed by the ECB Group, is expected to produce up to 20,000 barrels of HVO, Synthetic Paraffinic Kerosene (SPK), and Green Naphtha per day [116]. This facility will operate using different feedstocks, like soybean oil, used cooking oil, and animal fats, and is expected to start production in 2022 [117].

2.10. Peru

The first Peruvian law addressing biodiesel production was Law N° 28,054, also called “The Law for Promoting the Biofuel Market”. This law, released in August 2003, introduced definitions, goals, incentives and responsibilities on biofuel production and commercialization in Peru, specifically mentioning future biodiesel mixtures in their commercial class diesel [118]. In 2005, Supreme Decree N° 013/2005 regulated and further detailed the application of the Law N° 28,054, introducing 5% biodiesel fractions in N° 1 and N° 2 commercial diesel fuels, starting in 2010 [119]. This fraction was soon postponed via a new decree, setting biodiesel percentages to 2% in 2009 and 5% in 2011 [120].

From early on, the Peruvian biodiesel industry faced several problems, and was not able to become competitive enough to sustain national demand. One of the major barriers for the development of this sector in Peru was foreign competition, with dumping¹ episodes being reported by national producers. The first one was related to biodiesel imports from the United States (2009), which resulted in anti-dumping compensation measures being taken for biodiesel coming from the North-American country [121]. This opened pathways for Argentina, another great biofuel exporter, to dominate the national market, in such a way that Argentinean biodiesel came to represent 100% of all Peruvian imports in 2014 [122] and led local producers to completely stop production in the same year [123].

However, in the same period, new dumping accusations were raised,

¹ In economics, *dumping* is the practice of exporting goods with very low prices, representing an unfair competition to the companies in the importing country.

this time against Argentina, leading to new anti-dumping duties [123]. Despite this new hinder to the biodiesel importers, the Peruvian biodiesel producers were still unable to compete with other countries, and the reduction of Argentinean imports led fuel distributors to start importing from Indonesia and the Netherlands instead of being supplied by the local market [122]. With these issues, the Peruvian biodiesel industry only resumed production in 2017, probably as a late effect of anti-dumping duties formerly applied to major exporters. The production in this year, however, represented only 11% of national biodiesel consumption, which was expected to grow to 18% in 2018 [122]. Currently, Peruvian biodiesel production on a commercial scale is based on palm oil only, without participation of other oilseeds, like soybean and maize [124].

The difficulties faced by the Peruvian biodiesel industry can be observed in the national balance (Fig. 5). Similar to Colombia, no exports were reported, but Peru was able to keep internal consumption relatively stable via imports. Actually, imports were responsible to supply almost all Peruvian biodiesel demand up to 2016, when national production resumed and demand for foreign biofuels decreased. As was already mentioned, dumping episodes led the recently created national biodiesel industry to decrease or even completely stop production for several years, and dumping compensations took years to take effect. Even after the recovery of its biodiesel industry, however, Peru still depends on imports to supply its biodiesel demand.

2.11. Suriname

Suriname is the smallest South American country, with a total land area of 160,000 km² mostly covered by forest. It has a relatively small population density, which is higher at coastal regions and significantly lower inland [125]. Cultivated areas are mostly dominated by annual crops (mainly rice), with smaller fractions being divided between permanent crops like banana, sugarcane, coconut, citrus and palm oil tree plantations [126].

Similar to its neighboring territory, French Guiana, Suriname has a notable inequality between coastal and inland areas, and the latter is mostly energetically supplied by diesel generators [127]. Unlike French Guiana, however, Suriname presents a significant oil production and has its own refinery, which, alongside its high hydropower potential, allows it to be generally self-sustainable in terms of energy consumption [128].

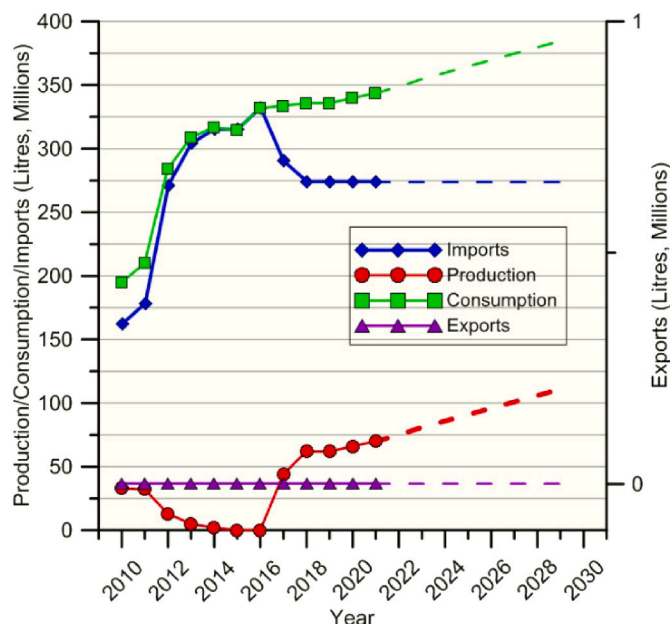


Fig. 5. Annual balance of the biodiesel industry in Peru [16].

Since oil production is enough to supply the country's demand and since most of the country's area is covered by forests, no biodiesel production is expected in the near future. In fact, a study on possible future pathways for the Surinamese energy sector only mentions using 1st and 2nd generation biofuels in one of the four considered scenarios, and no specific mention was made regarding biodiesel production [129].

2.12. Uruguay

In 2002, Uruguay published its first law encouraging biodiesel production in the country, defining entities that would be responsible for assessing viability, requirements and legal regulation regarding the introduction of this biofuel into the national market. This same law also allowed the government to promote a tax-free policy for any alternative fuel produced from Uruguayan animal or vegetable-based feedstock [130]. This biodiesel promotion policy was further specified by Law N° 18.195, published in 2007, that defined mixtures of up to 2% biodiesel in commercial diesel, that would gradually be increased to a mandatory minimum of 2% in 2009 and 5% in 2012. It is important to highlight that, different from other similar laws in South-American countries, Law N° 18.195 specified that biodiesel should be "produced in the country using national feedstock" [131], showing a concern for the stability of local producers.

Uruguayan biodiesel is produced by the state company ALUR S.A., which has two units totaling a production capacity of 83 million liters/year. These facilities use vegetable oil (soybeans and canola), animal fats (beef tallow) and recycled feedstock (waste frying oil) [132]. After production, this biodiesel is mixed with commercial diesel in correct proportions by the National Administration of Fuels, Alcohol and Portland Cement (ANCAP), also an Uruguayan state company to which ALUR S.A. belongs [133].

Regarding the production of new biofuels such as HVO, Uruguay does not currently have any official ongoing projects, but the recent Law N° 19.924, released in December 2020, extended Law N° 18.195 to also include "every renewable liquid fuel obtainable either from agricultural feedstocks or from industrial, agricultural or urban solid waste", potentially opening pathways for future HVO production in the country [134]. Following this law and the alleged interest of potential producers, the Uruguayan Ministry of Industry, Energy and Mines approved another resolution, also in December 2020, seeking to "boost (...) the implementation of biorefineries for the production of green diesel and jet fuel obtained from catalytic hydrogenation of oils and fats from vegetable and animal sources" [135]. Moreover, the viability of producing HVO in the country was already mentioned by government sources [136], and an early study carried out by the *Universidad de la República* found out that producing green diesel from tallow in Uruguay could be both technically and economically viable [137].

2.13. Venezuela

Venezuela has undergone serious political and economic crises in recent decades [138]. While Venezuela faces political crises, the country is extremely dependent on oil exports (it is the only South American country member of OPEC, owning one of the largest proven oil reserves in the world [54]), and has been suffering with falling oil prices [139]. Additionally, the country has basic sanitation problems, with lacks in water supply for 82% of the population according to a report presented by some Venezuelan NGOs [140], and hyperinflation was estimated at almost 20,000% in 2019 [141].

Since Venezuela is extremely dependent on fossil fuels and does not have regulatory or policy frameworks related to climate change, the country has not made many efforts to open up the possibility of renewable energy alternatives [142,143]. The Venezuelan government itself has already argued that biofuel production could be detrimental to food production [144]. Moreover, the Venezuelan government has a monopoly on electricity generation, resulting in lacks in investments in

alternative energy sources in the country [145,146].

In this sense, there is only incipient biodiesel production in Venezuela, especially in academic terms at universities [143,147,148]. Ogunkule and Ahmed [149], pointed out only one scientific publication in a journal indexed in Web of Science from Venezuela, which investigates the use of biodiesel in diesel engines. Therefore, there is no forecast for large-scale biodiesel production in the near future in Venezuela. Despite this, Bautista et al. [145] and Cremonese et al. [2] suggest future biodiesel production from second generation routes in Venezuela, avoiding problems related to fuel competition with food cultivation.

3. Analysis

The success or failure of implementing biodiesel in a country may depend on several factors, including feedstock availability and governmental willingness to respond to market changes as some of the most important variables for ensuring good and stable outcomes. Assessing different countries, with different natural resources, demands and policies offers a rich opportunity for understanding the main factors that need to be addressed in order to achieve a well-established biodiesel industry. Following, the main parameters that define the results of the biodiesel industry are discussed separately.

3.1. Early biodiesel regulations

As can be noticed by looking at real cases for South American countries, the release date for the first policies that deal with regulation and incentives for biodiesel production may indicate how mature a given national industry is. This, however, is not the only factor when defining a well-developed biodiesel market, since it also depends on several other variables. Table 1 shows that the long time since the first policies were released certainly helped countries like Brazil, Uruguay and Argentina reach stable biodiesel markets, whereas late regulation and incentives in Bolivia resulted in only recent discussions on implementing national production chains.

The case of Bolivia emphasizes the obvious assumption that an alternative fuel does not simply appear in a country's fuel matrix without a proper government regulation. The Bolivian soybean market is well-established and solid, but high feedstock availability could not ensure an industrial-scale biodiesel production, since related regulations

Table 1

Relation of early biodiesel policies and stage of development of the biodiesel industry in South American countries.

Country	First policy	Stage of development
Argentina	2006	Well-developed
Bolivia	2019	Early stage
Brazil	2003	Well-developed
Chile	2007	Developing
Colombia	2004	Developing
Ecuador	2004	Developing
French Guiana	–	Non-existent
Guyana	–	Non-existent
Paraguay	2005	Developing
Peru	2003	Developing
Suriname	–	Non-existent
Uruguay	2002	Well-developed
Venezuela	–	Non-existent

Well-developed: industries could respond to market variations and there is potential for future expansions.

Developing: industries could produce certain amounts of biodiesel, but countries were vulnerable to demand increases and/or foreign competition.

Early stage: industries only recently started production, or are expected to start in the near future.

Non-existent: no significant policies regarding biodiesel production have been published, and no large scale national production is expected in the near future.

only came out in 2019. When compared to its neighbors Brazil and Argentina, which also present high soybean production, the main difference that may justify the non-development of the Bolivian biodiesel industry is the time when first policies were implemented. In fact, there is a time gap of over 10 years between the first biodiesel regulations in Brazil and Argentina compared to Bolivia.

In general, policies should also be solid and able to quickly respond to variations in the market. Good examples are Argentina, Brazil and Uruguay, that, despite recent setbacks due to the coronavirus pandemic, were able to ensure the development of national biodiesel producing industries. The maturity, use of an “authority of application” and ability to dynamically respond to internal and external variations make the biodiesel policy in these three countries comparable to other leading biodiesel producers such as the United States, whose Environmental Protection Agency (EPA) actively controls the biofuels market in the country [150]. Important that early policies can be in defining the success of national biodiesel production, it is obviously not the only factor, which can be noticed by looking at cases such as Colombia, Ecuador, Paraguay and Peru. These countries released their first biodiesel policies in 2005 or before, yet were unable to reach stable national production. Following, factors other than early regulation implementation are discussed in order to understand such cases.

3.2. Soybean production and the success of national biodiesel industries

Even though soybean has a lower biodiesel yield per hectare compared to other oilseeds like sunflower, jatropha, castor and rapeseed [12], it seems to play a major role on the biodiesel production in several South American countries. Moreover, despite the aforementioned lower yield, soybean industry solidity seems to be directly related to the success of national biodiesel industries, as shown in Table 2.

In fact, in countries with well-developed biodiesel industry, like Argentina, Brazil and Uruguay, soybean represents more than 90% of all produced oilseeds. Even being an exception regarding the state of development of its biodiesel industry, Bolivia also used its high soybean production to justify recent advances towards the implementation of this biofuel production. At the same time, soybean production in Peru is nearly negligible and local biodiesel is commonly produced from palm oil, which may justify the difficulties faced by its industry against market volatility and Argentinean competition. Similar unsuccessful cases for palm oil based biodiesel can also be observed in Colombia and Ecuador, since this feedstock does not have a market as solid as soybean and has environmental issues that can hinder exports.

Finally, the tight association between soybean industry and big companies and landowners, alongside its well-established market due to its many by-products, may have given additional stability to biodiesel industries that chose to rely on this abundant feedstock. On one hand, the dominance of big producers in the soybean-biodiesel industry

Table 2
Soybean production and percentual participation in South American markets [151].

Country	2018 production in tonnes		Soybeans participation (%)
	Soybeans	Total oilseeds	
Argentina	37,787,927	42,930,057	88.02%
Bolivia	2,942,131	3,173,935	92.70%
Brazil	117,887,672	124,685,900	94.55%
Chile	NA	341,228	NA
Colombia	77,000	2,208,474	3.49%
Ecuador	25,504	760,148	3.36%
French Guiana	NA	2407	NA
Guiana	NA	136,876	NA
Paraguay	11,045,971	11,264,622	98.06%
Peru	1530	524,078	0.29%
Suriname	7	14,830	0.05%
Uruguay	1,334,000	1,437,017	92.83%
Venezuela	6000	235,225	2.55%

hinders social advancements like familiar agriculture, which was included as a goal in several countries' biodiesel regulations, but it seems to provide a strong stability to the sector and contribute to the development of economically solid biodiesel industries.

3.3. Foreign competition

Even when early policies are implemented and feedstock is available for biodiesel production, markets always seek out lower prices and better trade conditions. In this sense, foreign competition poses an additional barrier for developing biodiesel industries. The most evident case when it comes to market competition is, as already discussed, Peru. Despite having released its first biodiesel regulation policy in 2003, Peru has faced several setbacks in its national industry, postponing the increment of biodiesel in the fuel matrix, and even completely stopping national biodiesel production for some years, in light of competition from North-American and Argentinean producers. In this case, even anti-dumping compensations were not able to ensure immediate stability for local producers.

It is true that the fact that Peru bases its biodiesel production on palm-oil instead of soybean can also justify the difficulties faced by its national industry, as was already discussed, especially because one of the main competitors was its neighbor, Argentina, which has large-scale soybean-based production. The assessment of the Peruvian case, however, also emphasizes the role played by foreign market competition, showing that political regulations should ensure an additional protection for local producers to avoid similar outcomes. One example is Uruguay, with a well established biodiesel industry, whose Law N° 18.195 specifies that biodiesel should be produced in the country using national feedstock.

3.4. Technological availability

Even though it does not contain any developed countries, South America is mostly capable of producing biodiesel at an industrial scale, technologically speaking. The technology factor is not an issue for most countries in the region, but is particularly important for smaller ones. Even though it could pose a barrier for local production in French Guiana and Suriname, these nations do not seem to have any significant interest in investing in biodiesel production. In Guyana, however, the lack of technology is being overcome by partnerships with Brazil and other partners that have well-developed industries. For all other South American countries, however, other previously discussed factors seem to have had a much greater effect on defining the success or failure of national biodiesel industries.

3.5. Imminent competition with HVO

It is evident that many South American countries have not reached a reliable stable national biodiesel industry. Until now, market competition has been mainly concentrated on foreign biodiesel producers and on the fossil diesel itself. With respect to the latter, environmental aspects and related regulations weigh in favor of biodiesel, which make possible for this biofuel to be present in the market even when its prices exceed those of the traditional diesel. This scenario, however, is about to become harder for biodiesel producers, since a new biofuel is to be introduced in the South American market. The so-called “Green diesel” or Hydrotreated Vegetable Oil (HVO) was already successfully tested in Brazil, with the production of 40 million liters of co-processed 95% diesel-5% HVO [51], and industrial scale production is expected to start soon in Paraguay. Far from being simply “one more option”, HVO may surpass biodiesel in technical [152] and environmental [153] aspects, and its production process can also yield aviation fuel [154]. Moreover, since HVO can be produced from a variety of vegetable oils [155], a competition for primary resources may also occur.

Compared to diesel and biodiesel, HVO fuel produced by NESTE

(called NEXBTL™), has a much higher cetane number, reaching values between 70 and 95 [156], against 45 to 65 observed for biodiesel [157] and 37 to 56 for diesel [158]. Some studies highlight the benefits of using HVO in Diesel engines, like reduced CO, HC and NO_x emissions, as well as lower fuel consumption. These technical advantages were reported in several studies and for different engine configurations, as summarized in Table 3.

Despite the aforementioned benefits, HVO may face short-term problems for market entry and for replacing biodiesel in Latin American markets. In Brazil, for example, HVO costs may exceed the price of commercial diesel by more than three times, and by 65% the price of pure biodiesel [163]. HVO costs are directly associated to its intensive hydrogen consumption during the hydrogenation of triglycerides required in its production process [164], which also varies according to the level of saturation of the feedstock [165]. Therefore, as HVO is implemented in South American countries in the next years, its prices should strongly depend on the type of feedstock, level of technology and policy incentives.

Regardless of its prices and dependence of incoming policies, the fact that HVO is a promising biofuel that can easily adapt to feedstock availability while offering advantages over both diesel and biodiesel indicates that this new fuel will most probably find its market share in the near future. This means that industries that currently dominate the CI engine fuel market are already preparing for this imminent competition. Most of these possible reactions, however, are limited to recent meetings and unofficial institutional declarations.

In a meeting carried out in July 2020, Ubrabio, which represents biodiesel and biokerosene industries in Brazil, teamed up with representatives of the aviation industry to defend the integration of biodiesel, HVO and fossil diesel. Since HVO and biokerosene can be coproduced by the same processes [166], Ubrabio defended that future regulatory marks for “Green diesel” should be tied to aviation biofuel, while also arguing that HVO should replace a portion of the fossil diesel in a ternary commercial diesel containing HVO, diesel and biodiesel [167]. This, of course, defends both the biokerosene and biodiesel industries, since it ensures new sources for biokerosene production and keeps the market share occupied by biodiesel fuel intact. In a webinar carried out in 2020, however, state company Petrobras defended that co-processed HVO produced in Brazil could be mixed with additional biodiesel to meet the biofuel fraction in commercial diesel, as required by national regulations. This, of course, means that the 5% HVO in the mixture would partially replace biodiesel and not fossil diesel in final blends [52].

Table 3
Emission and performance effects from using HVO fuel in diesel engines.

Ref.	Type of analysis	Effects of HVO fuel
[152]	Performance and emissions of a passenger vehicle operating with 30–70% HVO blended with diesel	The addition of HVO to the blend reduced fuel consumption and also reduced CO, CO ₂ , HC, PM and NO _x emissions
[153]	Emission analysis in blends of RME with diesel and HVO	Reduced CO, HC and PM emission in blends that contained HVO as a replacement for traditional diesel
[159]	Performance and emissions of a light duty diesel engine operating with biodiesel and HVO blends	Blends containing HVO as a substitute for biodiesel had lower THC and CO emissions and had better cycle efficiency, and lower fuel consumption
[160]	Use of HVO and biodiesel from frying oil in heavy-duty trucks	Relevant reduction in CO ₂ and NO _x emissions when using HVO compared to biodiesel
[161]	Combustion and emissions, for diesel, biodiesel, HVO and their blends	Lower CO, HC and NO _x emissions and lower fuel consumption for HVO relative to biodiesel
[162]	Euro6 car operating with HVO and biodiesel	Shorter ignition delay, lower fuel consumption, lower CO ₂ and N ₂ O emissions and practically zero methane emissions for HVO consumption

At the same time, South American countries have started preparing for the imminent use of HVO. While a large-scale project is under way in Paraguay [116], Bolivia also announced a similar idea for producing this biofuel locally [9]. Even without any official project, Uruguay also gave signs to be interested in HVO production by promoting recent incentive policies and updating national biofuel regulations [134,135]. Other than that, case studies were also carried out in Colombia [76], and Uruguay [137], indicating that green diesel production could be both technically and economically viable.

It is still uncertain whether countries will create specific regulations for HVO or just extend the ones that already cover biodiesel production. Also, the fact that the facilities designed for producing this new biofuel could be either independent (as in Paraguay [116]) or coupled to existing biodiesel plants (as in Bolivia [9]) puts doubt on whether this new market will be dominated by new companies or by biodiesel producers already transitioning towards more modern fuels. Finally, since HVO could replace both diesel and biodiesel in technical terms, the impact that this biofuel could cause in the biodiesel industry is still not clear. It is likely, however, that in case the ongoing projects succeed in producing high-quality, economically viable HVO, other South American countries will follow the same path, taking advantage of existing logistics networks and feedstock availability to accelerate the transition towards a cleaner and more modern fuel matrix.

3.6. Transition towards an electrified transportation

In the last few years, regulations regarding the use of electrical vehicles and even about the extinction of combustion engines have started to emerge worldwide. Many countries like Canada, Japan, Mexico and several European countries have already announced plans for phasing out combustion engines for transportation [8,168]. The European Union signaled the end of vehicles that produce carbon emissions by 2035 [169], and several U.S. states urged the federal government to follow the same path [170]. In South America, a yet to be passed bill for banning fossil-based internal combustion engines by 2030 was proposed in the Brazilian Senate [171], and the Argentinean government recently presented a similar project [172]. In this sense, the political scenario and constant pressure for this supposedly greener solution indicate that a large-scale transition towards the electrification of the transportation matrix may actually occur in the next years; yet, this is not a consensus, and the role of biofuels can also be presented as a strong alternative for reducing GHG emissions while total electrification is not fully viable [8].

Discussing the possible electrification of the transportation sector, Senecal and Leach [173] published a book regarding the balanced approach for this sector. The authors highlighted the importance of an eclectic future between combustion and electrification and that all technologies, from internal combustion engines to battery electric vehicles, cause environmental impacts. The electrification may be the right path towards a sustainable future in transportation for some regions. However, in developing areas there is no electric infrastructure or favorable energy matrix to sustain the vehicles electrification. Also, the majority of vehicles today use the internal combustion engine, and this is unlikely to change anytime soon [174]. In this sense, improvements in engines and development of biofuels, together with hybridization, seem to be the way forward, especially for South America [175].

Additionally, Santos et al. [176] stated that, despite electric vehicles have been promoted by some parties as the only and best solution for the sustainable future of transportation, a complete disruption of current transportation technology, in this case using internal combustion engines, is not desirable. The authors demonstrated the risks of internal combustion engine banment, especially in South America region, and the importance of biofuels in economic, environmental and political scenarios. A lack of regulation and national consensus in South American countries about the implementation of electric vehicles adds to the drawbacks and reinforces the long-term perspectives for electrification [177].

Whether vehicle electrification will be a reality soon or not, this is a challenge to be considered by biodiesel producers. The strong dependence of this possible change to political aspects make this scenario even more uncertain, particularly for biodiesel exporting countries like Argentina and Brazil, in such a way that biodiesel producers should be able to respond to changes in both internal and external markets. The possible electrification of the transportation sector is a double-edged sword for biodiesel and other biofuels producers: while it could represent the end of internal combustion engines in general, it could also open pathways for greener fuels, which are able to replace fossil diesel while also decreasing GHG emissions, ultimately favoring biofuel producers in biomass producing regions like South America. In this sense, at least when it comes to internal consumption in South America, the latter seems to be much more likely to occur in the near future, given the high biomass availability, political influence of regional crop producers and existence of already developed biofuel industries.

4. Conclusions

Over the last years, both population and energy demand in South American countries have increased. In order to supply this growing demand, installed capacity was increased and research on new and existing sources of energy was constant. In the transportation sector, partially due to the major role played by agriculture in many South American economies, biofuels like biodiesel have been widely used in the last few decades. While some countries seem to be only starting the implementation of this biofuel in their transportation matrix, others already rely on it to supply the current energy demand and seem to be even moving towards the introduction of other biomass-based fuels.

As expected for any energy-related product, biodiesel viability strongly depends on local policies, which may either facilitate or harden its implementation in a given market. The prosperity of biodiesel success in South American countries was also influenced by other factors, such as competition with other markets or fuels and delays in implementing incentive policies for supporting the establishment of this biofuel at national level. Over time, some countries like Peru and Paraguay, have proposed goals and policies for biodiesel implementation that could not be achieved in due time, and ended up being either postponed or replaced by more mild targets.

The Brazilian biodiesel industry can be considered a case of success due to its high production, intense participation in the market and growth potential, but has failed in promoting social inclusion by means of family agriculture in less favored regions. In terms of stability and economic viability, Argentina and Uruguay can also be placed among the cases of success, with early regulations and a solid soybean-based biodiesel industry. Delay in regulations was certainly the most important factor in the non-development of the biodiesel industry in Bolivia, which has high feedstock availability due to its high soybean production. Despite any social issues related to the dominance of big producers in the soybean market, this feedstock seems to be strongly associated with the success of large-scale biodiesel production, contrary to palm based production, which was not able to ensure similar stability in Peruvian, Colombian and Ecuadorian industries.

Competition with international market has also posed a barrier for progress of local biodiesel industries in Peru, with dumping episodes resulting in long judicial disputes and prejudicing the development of the fragile starting industries. This hindered national production to the point that production was completely stopped for several years, and setbacks were observed in some regulations. The lack of technology for biodiesel production does not seem to have a major impact in South American biodiesel production, with Guyana being the only country to be associated to such difficulty. Other smaller countries and territory, like Suriname and French Guiana, does not seem to be interested in local biodiesel production despite any lack of technology.

Finally, the next challenge to be faced by the biodiesel industry in South America is the imminent use of HVO. This fuel has already been

co-processed with fossil diesel in a pilot plant in Brazil, and large scale production is expected to start in Paraguay and Bolivia in the next years. Uruguay recently adapted national regulations to open pathways for producing HVO locally, and case studies indicated the viability of similar projects in Colombia and Uruguay. Since this new biofuel can replace both diesel and biodiesel, its impact over the biodiesel industry is still not clear, and will strongly depend on the focus of new policies and the ability of producers to properly react to this new competition. Similarly, the possible electrification of vehicles in both South America and in foreign biodiesel importing countries increases the uncertainty for local biofuel producers, since the future of these fuels and combustion engines in general will strongly depend on political, geographic and economic aspects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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